

Trade, quality and pass-through: The U.S.-China solar panel dispute

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Abstract

The 2018 U.S.-China trade war has received a lot of attention for its impact on the term of trade and overall was evaluated to have a small negative impact on U.S. consumers. However, some dimensions of this trade war are left untouched, noticeably the importance of quality of the traded goods in assessing the welfare effects of the tariffs. We base our analysis on a heterogeneous firms model including endogenous markups and quality. Leveraging an installed solar system level database, we can access the importance of quality in the terms of trade. We found **re-sults**, inline/not inline with the model predictions due to **explanation**

Keywords: Trade, Quality, Tariffs, Heterogeneous Firms, Endogeneous Markups, Endogeneous Quality.

1 Introduction

2 Theory

In this section, we develop the environment in which we assume the solar market evolves and that motivates our empirical analysis.

For this purpose, we adopt the ? framework, prolonged by ? and ? to integrate endogenous quality choice and disentangle the quality effect impact on pass-through. We prolonged this model by adding a third country unaffected by the newly imposed tariffs. The interest is both theoretical in assessing the welfare repartition when we include a third country and empirical in that the U.S. solar panel market is largely dominated by three countries of origin brands: the United States, China, and South Korea.

2.1 Demand

The world is divided into two countries: Home (H) and Foreign (F). For simplicity, we assume that both countries have a representative consumer with a quasi-linear utility function. This function is largely borrowed from ? and was developed in trade by ?.

$$U = q_0^c + \int_{i \in \Omega} (\alpha + z_i) q_i^c di - \frac{1}{2} \gamma \int_{i \in \Omega} (q_i^c)^2 di - \frac{1}{2} \eta \left(\int_{i \in \Omega} q_i^c \right)^2 \quad (1)$$

We have a q_0^c the numeraire good which is homogeneous and produced by a unit of labor and is always consumed. The presence of a numeraire in our setup gives a partial equilibrium flavor to our model, since wages will not vary with trade and firms always have enough workers to deliver to all markets. In our case this partial equilibrium effect is not necessarily problematic since we only consider the specific case of tariff absorption in solar panel trade with the U.S. We also have a differentiated good q_i^c over which consumers maximize their utility. The differentiated good is indexed by variety $i \in \Omega$ and each firm produces a unique variety of this good (in our specific case solar panels). α and η are strictly positive and capture the substitutability between the numeraire and the differentiated good, while γ captures the degree of horizontal differentiation between varieties. z_i is the quality scaling factor.

Using quasi-linear utility functions has the double convenience of providing a tractable framework for firms' heterogeneity while allowing for markups and quality to be determined endogenously at the firm level. The classical CES function posits constant elasticity of substitution and then constant markups which can hide strategic patterns of tariffs absorption which are precious in our context.

To Do

- why did we choose it over CES/CREMR/translog
- provide the "demand manifold" ?
- explaining why choosing quasi linear and caveats (??)

We assume that the two markets are segmented and derive the following inverse demand function:

$$p_i^l = \alpha + z_i^l - \gamma(q_i^c)^l - \eta Q_c^l \quad (2)$$

with $l \in \{H, F\}$ and $Q_c^l = \int_{i \in \Omega} (q_i^c)^l di$

From here we can derive the linear demand for any variety in a country l by just inverting the function :

$$q^l \equiv L^l q_i^c = \frac{L^l \alpha}{\gamma + \eta N^l} + \frac{L^l}{\gamma} z_i^l - \frac{L^l}{\gamma} p_i^l - \frac{L^l N^l \eta}{\gamma + \eta N^l} \bar{z}^l + \frac{L^l N^l \eta}{\gamma + \eta N^l} \bar{p}^l \quad (3)$$

2.2 Firms

We assume that there is a continuum of firms N^l operating in each country and that those firms are in monopolistic competitions. They can endogenously set markups and quality, and take decisions independently of each other. **describe results from ?**

They all face an entry cost in the market determined by f_E and produce a single variety.

Thanks to monopolistic competition property and the inelastic labor supply, they can deal independently with each market setting markups and quality of goods to export **cite papers going in that direction.**

We model the firms costs as follow :

$$TC_i^l = c_i q_i^l + \theta z_i^l q_i^l + \theta (z_i^l)^2 \quad (4)$$

$c_i q_i^l$ marks here the raising marginal cost c with the quantity produced, it is an output dependent "processing cost". $\theta z_i^l q_i^l$ refers here to the additional cost of producing quality goods and $\theta (z_i^l)^2$ is the design or research cost of upgrading. θ here refers to the "ability" to upgrade, in ? it is understood at the country level as the high and low innovation countries. These different elements integrated in the cost function allow firms to vary in quality and quantity by destination.

Firms ignore their productivity before entering the market; it is revealed when they pick a marginal cost of production c in a distribution $G(c)$. Entering

a market comes with a fixed cost of entry f_E , which is considered sunk here since the model is static. If the expected profit π of a firm is revealed to be inferior to f_E then the firm stops producing and exits the market.

In our setup, we are in an open economy **at the equilibrium**, firms make profit at Home (H) and at Foreign (F).

$$\begin{aligned}
\pi^{hh} &= p^{hh} q^{hh} - c q^{hh} - \theta z^{hh} q^{hh} - \theta (z^{hh})^2 \\
\pi^{hf} &= \frac{p^{hf}}{\tau^f} q^{hf} - \delta_C q^{hf} - \theta z^{hf} q^{hf} - \theta (z^{hf})^2 \\
\pi^{ff} &= p^{ff} q^{ff} - c q^{ff} - \theta z^{ff} q^{ff} - \theta (z^{ff})^2 \\
\pi^{fh} &= \frac{p^{fh}}{\tau^h} q^{fh} - \delta_C q^{fh} - \theta z^{fh} q^{fh} - \theta (z^{fh})^2
\end{aligned} \tag{5}$$

With p^{hh} and q^{hh} respectively indicating price and quantity of firms producing at Home h and selling on the same market. Conversely, firms face trade costs when exporting τ^f , they are independent of quality¹, and are our variable of interest in our case of pass-through evaluation.

In our case, we are interested in the effect of Home imposing tariffs on Foreign exports to its domestic market. Therefore we focus on :

$$\begin{aligned}
\pi^{hh} &= p^{hh} q^{hh} - c q^{hh} - \theta z^{hh} q^{hh} - \theta (z^{hh})^2 \\
\pi^{fh} &= \frac{p^{fh}}{\tau^h} q^{fh} - \delta_C q^{fh} - \theta z^{fh} q^{fh} - \theta (z^{fh})^2
\end{aligned} \tag{6}$$

And more precisely the pass through of the tariffs on the tariff inclusive import price: $\frac{p^{fh}}{\tau^h}$.

Firms choose quality and price to deliver to each market independently to maximize their profits. The optimal quality (see

With $\lambda^{fh} = \frac{L^h(1-\theta\tau^h)}{4\gamma\theta\tau^h-L^h(1-\theta\tau^h)^2}$ and $\lambda^{hh} = \frac{L^h(1-\theta)}{4\gamma\theta-L^h(1-\theta)^2}$ ². λ is defined as the "quality scope" the ladder of quality on which an industry operates. A flat ladder ($\lambda = 1$) would mean that the industry produces homogeneous goods and there is not much room for differentiation. In that case, prices are declining with productivity and we are back to ???. Conversely, with a steep ladder ($\lambda > 1$), firms with higher productivity can set both higher quality and prices.

More generally, the quality scope is increasing in market size (L) and substitutability between differentiated goods (γ). On the other hand, it decreases

¹Here we choose to follow ? and impose trade cost included in the production price rather than iceberg cost. Also there is no reason to believe *a priori* that high quality solar panels are more expensive to trade than low quality ones since most of the technology we observe is on the chemical treatment of raw materials or semi-conductor manufacturing, rather than on the size.

²We assume that $4\theta\gamma > L^l(1-\theta\tau^l)^2$ and $1 > \theta\tau^l$ so that the scope is always positive.

with tariff (τ) and toughness to upgrade (θ). In a nutshell, a large market, with steep competition between differentiated goods and with a high ability to innovate (e.g. a *developed* country) would have more variation in the quality of goods proposed because a larger market means more possibility for firms to make up their upgrading cost. Conversely, a market with strong tariff τ is analogous to a shock in size (market size reduction) in this two-country framework, reducing the opportunity to pay back for quality upgrading cost, and then leading to a more homogeneous market.

From the firm perspective, the choice to deliver a certain quality to a given market is synthesized by this scope for quality (market-based parameter) and their relative marginal cost relative to the market cost-cutoff ($c^{fh} - c$). Since the scope for quality on a given market is set by the market conditions, the quality of the variety delivered to a given market is monotonically increasing with the firm productivity ($1/c$). This relation between cost differential and quality scope is crucial since the tariff will affect firms differently depending on their productivity level.

For foreign firms exporting to the domestic markets :

$$\begin{aligned}
z^{fh} &= \lambda^{fh} \delta \tau^h (c^{fh} - c) \\
p^{fh} &= \frac{1}{2} \delta \tau^h (c^{fh} + c) + \frac{1}{2} (1 + \tau^h \theta) \lambda^{fh} \delta \tau^h (c^{fh} - c) \\
q^{fh} &= \frac{L^h}{2\gamma} [\delta \tau^h (c^{fh} - c) + (1 - \tau^h \theta) \lambda^{fh} \delta \tau^h (c^{fh} - c)] \\
\pi^{fh} &= \frac{\delta^2 L^h \tau^h}{4\gamma} [1 + (1 - \theta \tau^h) \lambda^{fh}] (c^{fh} - c)^2
\end{aligned} \tag{7}$$

For domestic firms producing for the domestic market we obtain the following functions:

$$\begin{aligned}
z^{hh} &= \lambda^{hh} (c^{hh} - c) \\
p^{hh} &= \frac{1}{2} (c^{hh} + c) + \frac{1}{2} (1 + \theta) \lambda^{hh} (c^{hh} - c) \\
q^{hh} &= \frac{L^h}{2\gamma} [(c^{hh} - c) + (1 - \theta) \lambda^{hh} (c^{hh} - c)] \\
\pi^{hh} &= \frac{L^h}{4\gamma} [1 + (1 - \theta) \lambda^{hh}] (c^{hh} - c)^2
\end{aligned} \tag{8}$$

2.3 Market Equilibrium

To close the model we need equilibrium conditions under which firm can operate in a given market. For that they must have positive profit just so

to pay the fixed entry cost (f_E) of entering a given market. When a firm draws its marginal cost c if $c > c^{fh}$ then the firm is losing money and stops producing. It states that a firm must cover the fixed cost entry by its profit at home or abroad.

$$\begin{aligned} f_E &= \int_0^{c^{fh}} \pi^{fh} dG(c) + \int_0^{c^{ff}} \pi^{ff} dG(c) \\ f_E &= \int_0^{c^{hf}} \pi^{hf} dG(c) + \int_0^{c^{hh}} \pi^{hh} dG(c) \end{aligned} \quad (9)$$

From these equations we can determine the market cost-cutoff above which firms cannot operate profitably. As it is standard, we will assume that the marginal cost c is distributed Pareto such that $G(c) = (c/c_M)^k$ with a support $c \in [0, c_M]$ with c_M being the maximum value c can take and k the shape parameter. A higher k implies more concentration of firms on the right tail of the distribution. Using Pareto distribution is convenient as it is a scale free distribution with noticeably interesting properties for left truncation.

However, if the Pareto distribution is convenient due to its analytical tractability and its aggregate properties, it has been subject to critics by ? and support ?.

$$\begin{cases} \gamma\phi = L^h \delta^{-k} (\tau^h)^{-k-1} [1 + (1 - \theta\tau^h)\lambda^{fh}] (c^{hh})^{k+2} + L^f [1 + (1 - \theta)\lambda^{ff}] (c^{ff})^{k+2} \\ \gamma\phi = L^f \delta^{-k} (\tau^f)^{-k-1} [1 + (1 - \theta\tau^f)\lambda^{hf}] (c^{ff})^{k+2} + L^h [1 + (1 - \theta)\lambda^{hh}] (c^{hh})^{k+2} \end{cases}$$

With $\phi = c_M^k 2(k+2)(k+1)f_E$, $c^{fh} = \frac{c^{hh}}{\delta\tau^h}$. Then we have this system of two equations that we can build on to obtain the expression of the cost cutoff of h and f :

$$\begin{aligned} c^{hh} &= \left(\frac{\gamma\phi}{L^h} \frac{[1 + (1 - \theta)\lambda^{ff}] - \rho^f [1 + (1 - \theta\tau^f)\lambda^{hf}]}{[1 + (1 - \theta)\lambda^{hh}] [1 + (1 - \theta)\lambda^{ff}] - \rho^h \rho^f [1 + (1 - \theta\tau^h)\lambda^{fh}] [1 + (1 - \theta\tau^f)\lambda^{hf}]} \right)^{\frac{1}{k+2}} \\ c^{fh} &= \left(\frac{\gamma\phi}{\delta\tau^h L^h} \frac{[1 + (1 - \theta)\lambda^{ff}] - \rho^f [1 + (1 - \theta\tau^f)\lambda^{hf}]}{[1 + (1 - \theta)\lambda^{hh}] [1 + (1 - \theta)\lambda^{ff}] - \rho^h \rho^f [1 + (1 - \theta\tau^h)\lambda^{fh}] [1 + (1 - \theta\tau^f)\lambda^{hf}]} \right)^{\frac{1}{k+2}} \end{aligned} \quad (10)$$

Note that $\rho^l = \delta^{-k} (\tau^l)^{-k-1}$ with $l \in \{h, f\}$ is the inverse measure of economic freedom as in ?. These expressions provide closed forms solutions of the model and intuition on what are the dynamics at play. We can see that the import cutoff value (from *home* perspective) is given by c^{fh} . $\frac{\gamma\phi}{\delta\tau^h L^h}$ give intuition on home market profitability with $\gamma\phi$ being the objective measure of

market access conditions divided by the trade cost $\delta\tau^h$. L plays its competitive role as in ?, an increase in size leads to a lower cost-cutoff leading less productive firms to exit the market, increasing the productivity. However, the vertical quality differentiation changes the intuition on prices and markups which do not necessarily decrease with market size increases since firms producing high-quality variety can also impose a higher markup (see ??).

We now turn our analysis to the equilibrium import price structure, we can decompose it following ?? as being formed of two components productivity and quality:

$$p^* \equiv \frac{p^{fh}}{\tau^h} \implies p^* = \underbrace{\frac{\delta}{2}(c^{fh} + c)}_{\text{Productivity}} + \underbrace{\frac{\delta}{2}\kappa(c^{fh} - c)}_{\text{Quality}} \quad (11)$$

To alleviate the notation we have set $\kappa = (1 + \tau^h\theta)\lambda^{fh}$ which is a proportional measure of the quality scope for differentiation in the market. We now have an expression of import price determined by three key parameters: firm productivity c , market import cost-cutoff c^{fh} , and the quality scope for differentiation κ . Taking the total log derivative we have (full derivation in ??):

$$d\ln(p^*) = \frac{\delta}{2} \left[\underbrace{\frac{(1 + \kappa)c^{fh}d\ln(c^{fh})}{p^*}}_{\text{Market Condition}} - \underbrace{\frac{(1 - \kappa)cd\ln(\frac{1}{c})}{p^*}}_{\text{Productivity}} + \underbrace{\frac{(c^{fh} - c)\kappa d\ln(\kappa)}{p^*}}_{\text{Quality}} \right] \quad (12)$$

We can clearly see the different effects at play in the import price equilibrium, while trade cost $\frac{\delta}{2}$ mechanically increases the cost of import, the quality, productivity, and market conditions interplay in non-trivial ways.

First, depending on the value of κ the sign of productivity effect on price can change. If $\kappa < 1$ then productivity correlates negatively with prices, and more productive firms will set a lower price than less productive ones. This is the traditional pro-competitive effect of trade identified in ? which reduces markups and prices (if we omit the quality component). If $\kappa > 1$ then the sign of the productivity effect shift, and the correlation between productivity and prices appears to be positive (**quote studies that find this results**). This result, at first glance at odds with ?, comes from the quality scope of consumed goods. In that case, both quality and productivity would push up the prices. The logic is that more productive firms would also produce more high-quality goods (see (??)). Hence, quality adds ambiguity to the correlation between productivity and import/export prices. The sign of the correlation is an empirical question that we will tackle in ??.

Second, the cost cutoff is always positively correlated to the import prices, an increase in cost cutoff will make this market less competitive by allowing less productive firms to maintain themselves in the market. It will also create the possibility for more productive firms to charge higher markups, making the market more profitable. This situation then would decrease the average productivity and increase the prices and markups. A decrease in the cost cutoff would have the opposite effect.

Third, quality, when it plays, always tends to increase the prices of imports as higher-quality goods are more expensive.

Then overall effect of tariffs depends on the degree of goods homogeneity (following ? we define as good homogeneous goods which $\kappa < 1$ and quality differentiated goods which $\kappa > 1$), and the tariff absorption of firm exporting to *home*.

2.4 Tariffs Absorption

Now that we have our core setup with the equilibrium import price defined, we can study the elasticity of prices with respect to tariffs, which will give us the level of pass-through onto consumers that we should expect. Given the linear form of the demand, unlike in CES function, the pass-through can vary by firm. These non-constant markups allow for strategic pricing of firms depending on the elasticity of demand they face. The elasticity of demand faced by firms largely shapes their behavior and then transmission of the pass-through (??).

The tariff absorption elasticity is defined as (see derivation in ??):

$$\epsilon_{p^*} = -\frac{\partial \ln(p^*)}{\partial \ln \tau^h} \quad (13)$$

It measures the impact of 1% increase in tariff on home price in %. We can in (??) that the tariffs influence the price through the quality and cost-cutoff channels, so far productivity is left aside as tariffs are not directly impacting firms in our model³.

So far we assume that the result of the derivation is identical as in ?.

³We implicitly assume that the productivity of firms is unaffected because firms produce in their homeland f or h and are not internationalized or do not choose to invest in another country market (FDI) like in ?. It comes from the fact that our unique factor of production is labor and is assumed to be immutable.

To Do

- **We can also build on the theoretical work of "demand manifolds"** (?) and access from the residual firms faces their likely reaction. Our linear demand functions fall under the "manifold invariance" due to its "bipolar direct" family. It implies that the shape of "demand manifold" does not change, only the level given the parameter. Therefore, comparative statistics are easier at the equilibrium.
- "demand manifold" concept, check if we are in the context of bipolar function (i.e. : $P(x) = Ax^{-\alpha} + Bx^{-\beta}$) of parameter $\alpha = 0$ and $\beta = -1$
- Be critical about the use of Pareto-inversion price
- Include the theory bits and clarification from **Parenti & al (2017)**

2.5 Quality choice and quasi-Metzler paradox

3 Data

4 Empirical

In this section, we provide an empirical analysis based on the rationale of our model. Our main objective will be to assess the differentiated pass-through effect by firms given their quality product.

Questions

- For the market data: having a fixed effect for the installer and for the city of installation wealth?

1. **Result 1:** Evaluate the degree of homogeneity of our industry
2. **Result 2:** If our industry is homogeneous then we should see larger tariff absorption from high quality firm, if our industry heterogeneous we should see less tariffs absorption iff the cost of innovating is not too high.
3. **Result 3:** Discuss the quasi-Metzler paradox due to quality increase and propose a new way to estimate it.
4. **Result 4:** Estimate demand elasticity at the good level

4.1 Key Variables

?? reveals that there cannot be a unique criterion for quality based on efficiency given the rapid improvement of the industry. If we were to set an arbitrary efficiency threshold for quality at 20% we would be considering in 2011 close to 0 panels able to reach this level of efficiency. Therefore, we define the premium nature of the panel given its place in the distribution.

From here we can build a first set of quality criteria based on the *efficiency* of the solar panels model, but also on the presence of additional features such as an integrated micro-inverter which contributes to the overall quality of the product.

Quality Metrics 1 We described as *premium panels*, panels whose efficiency is in the top 5% of the efficiency distribution for the year preceding the implementation of the tariff. This is a dummy variable.

This approach has already been used in past work on this database (??). However, to reinforce our empirical approach and embrace the various notions of quality that guide consumer choices we adopt the following alternative metrics.

Quality Metrics 2 We define as *Premium Installations* the installation that possesses not only *premium panels* but also integrated micro-inverter and/or DC optimizer. This is a dummy variable.

Integrated micro-inverters and DC optimizers contribute to the overall quality of solar panel installations but also increase the upfront cost of such installation. Therefore, including them as an additional quality criterion also extends the scope of our quality analysis.

4.2 Good homogeneity

In coherence with our model, we want to test whether the solar panel market can be described as a homogeneous good market or not. The first way to test is by regressing the price on quality and see if there is a premium for quality. If ?? provides indirect evidence of quality scope, the regression provides a consumer-view perspective on the taste for quality. In this logic, higher quality products would be more expensive, which is also motivated by (??).

$$\ln p_i = \beta_1 \text{Quality} + \beta_2 \text{Country} \times \text{Quality} + \delta_s + \delta_t + \delta_i + \delta_m + \varepsilon_i \quad (14)$$

where p_i is the pre-incentive price (\$/W), *Quality* being either *premium panels* or *premium installations*, *Country* represent the country of origin of the brand⁴, and fixed effects for state (s), year (t), the installer i , and the manufacturer (m). The standard errors are clustered at zip code level.

⁴We do not observe the manufacturing place of each panel, here country refers to the branding of the firm as being "American" because of its HQ location, even though the production may be in Philippine or Malaysia.

Table 1: Price response to premium panels for HO and TPO systems

	HO				TPO			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Premium Panels	-0.285 (0.005)	-0.111 (0.005)	-0.051 (0.006)	-0.025 (0.005)	-0.578 (0.062)	-0.478 (0.062)	-0.457 (0.063)	-0.103 (0.040)
Premium Panels $\times$ Chinese Brand	0.077 (0.014)	0.040 (0.014)	-0.018 (0.013)	0.004 (0.010)	0.640 (0.064)	0.621 (0.063)	0.535 (0.065)	0.202 (0.041)
Premium Panels $\times$ Korean Brand	0.232 (0.007)	0.111 (0.007)	0.074 (0.008)	0.048 (0.006)	0.478 (0.062)	0.372 (0.061)	0.321 (0.063)	0.029 (0.041)
Premium Panels $\times$ USA Brand	0.113 (0.008)	0.138 (0.007)	0.040 (0.008)	0.068 (0.008)	0.224 (0.070)	0.286 (0.068)	0.252 (0.069)	-0.004 (0.041)
Num.Obs.	509 728	509 728	509 728	509 728	400 744	400 744	400 744	400 744
R2	0.087	0.238	0.245	0.491	0.087	0.214	0.239	0.576
FE: Year		Yes	Yes	Yes		Yes	Yes	Yes
FE: State		Yes	Yes	Yes		Yes	Yes	Yes
FE: Manufacturer			Yes	Yes			Yes	Yes
FE: Installer				Yes				Yes
F-test p-value	0.000	0.000	0.000	1.000	0.000	0.000	0.000	0.000
Wald-test p-value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Notes: The dependent variable is a log price per W, so the estimate reports percent variations in price. HO and TPO correspond to 'Host Owned' and 'Third Party Owned' systems. The standard errors are clustered at the zip code level shown between parenthesis.

?? provides evidence of the presence of a small premium for quality type 1 panels in both HO and TPO systems. The presence of fixed effects, however, plays an important role in our estimation strategy. Given the fast pace at which prices declined during the decade (see **Figure**) and state demand shock associated with a possible change in subsidy policy or local economic boom or bust, adding these fixed effects reduces by two the estimated effects of quality on price. The negative overall effects of quality on price are probably because of Chinese, Korean, and American brands, which account for more than 70% of the market shares, and panel varieties which absorb all the positive variation due to quality. We see that our quality premiums are strong and significant for Korean and American brands, which also correspond to qualitative insights from industry professionals (?). It oscillates between 7.4% and 4.8% for Korean brands, and 4% to 6.8% for American brands depending on the inclusion of installer fixed effects, and is not statistically different from 0 for Chinese brands.

When turning to the TPO systems we find very strong evidence of premium pricing though only statistically significant for Chinese brands, around 20%. This surprising result might indicate that Chinese brands might be more distributed among third-party owner companies see **reg**. If we are to believe

this correlation, this last result is coherent with the fact that homeowners not buying through intermediary providers do not provide a premium to Chinese panels, which can then be captured by TPO providers. This last result is in line with both qualitative (?) and academic evidence (?).

However, these descriptive results have a few caveat. The first is that for HO system once we introduce a fixed effect for installer, even though our estimate are individually significant, we massively reject the joint significance with the F-test. All the variation seems to be absorbed by the installer fixed effects, which reflects both the great dispersion in prices but also of installation. If very few firms produce 90% of the panel on the market, there are more than 5800 installers. The distribution of installation is very different in the two systems types (see ??). If the mean installation per installer is the same for both system, we can see in ?? that the distribution for TPO system is much more narrow and concentrated among large actors while for HO system, if there are big actors, overall the installation system is more spread. This reflect two logic of buying. One with an installer buying large quantity of (Chinese) solar panels largely passing the cost onto consumers, and the other consumers buying panels and contracting with local installers. Therefore, it considerably reduces the variation once it is captures by the fixed effects for HO systems. Conversely, if we do not observe a similar result for TPO system it is probably because fewer installer assure a greater share of the market. Finally, the Wald-test are significative in all configurations which reassures us on the presence of quality markups.

The second, is that we consider a threshold of 20% of efficiency over the period for this first quality metric, which represent top 5% efficiency only after 2013. It covers 7 out of 10 years of our sample but is a limitation of our analysis for the first year of the period considered. Nonetheless, the number of installations quickly increased after 2013⁵, making it a negligible issue here. For completeness, we included a period relative quality metrics following our definition (see).

⁵90% of the observations are after 2013

Table 2: Price response to premium installations for HO and TPO systems

	HO				TPO			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Premium Installations	-0.076 (0.005)	-0.017 (0.005)	-0.100 (0.017)	-0.017 (0.020)	-0.206 (0.029)	-0.334 (0.020)	-0.354 (0.022)	-0.127 (0.013)
Premium Installations $\times$ Chinese Brand	0.006 (0.008)	0.107 (0.007)	0.174 (0.018)	0.049 (0.021)	0.059 (0.029)	0.390 (0.019)	0.407 (0.022)	0.203 (0.012)
Premium Installations $\times$ Korean Brand	0.011 (0.007)	-0.019 (0.007)	0.066 (0.017)	0.006 (0.020)	0.259 (0.031)	0.412 (0.022)	0.462 (0.023)	0.234 (0.015)
Premium Installations $\times$ USA Brand	0.119 (0.012)	0.078 (0.010)	0.151 (0.018)	0.071 (0.021)	0.368 (0.066)	0.493 (0.063)	0.591 (0.067)	0.187 (0.019)
Num.Obs.	509 728	509 728	509 728	509 728	400 744	400 744	400 744	400 744
R2	0.053	0.237	0.248	0.491	0.093	0.219	0.242	0.578
FE: Year		Yes	Yes	Yes		Yes	Yes	Yes
FE: State		Yes	Yes	Yes		Yes	Yes	Yes
FE: Manufacturer			Yes	Yes			Yes	Yes
FE: Installer				Yes				Yes
F-test p-value	0.000	0.000	0.000	1.000	0.000	0.000	0.000	0.000
Wald-test p-value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Notes: The dependent variable is a log price per W, so the estimate reports percent variations in price. HO and TPO correspond to 'Host Owned' and 'Third Party Owned' systems. The standard errors are clustered at the zip code level shown between parenthesis.

Our second quality metrics bring roughly similar estimates of quality premium for Chinese, Korean and American brands for HO systems with the full set of fixed effects. Chinese and American brands benefit of 6% and 3.5% premium on prices, but Korean brands are now not statistically different from 0. Moreover, we can notice that premium are estimated to be much higher overall with this configuration, being 2 digits in most configurations. This even more striking when shifting to TPO systems. All firms have large premium markups, close to or above 20%. This might reflect two things with respect to our first quality metrics. First, we include more panels than just the one with highest module efficiency of the distribution, less efficient panels but with premium features (i.e. built-in micro inverter) are sold for a relatively higher price as they improve the overall efficiency of the installation. For HO system it reflects with slightly higher premium than in ???. But for TPO systems, it reflects very high markups, meaning they are probably able to bargain more with their customers or by default install more complete systems to pass-on more costs. If those results seems high, they are in line with other estimations such as ?.

4.3 Tariff Pass Through

Table 3: Caption

We are at the product level :

$$\ln p^* = \beta_1 \text{tariff} + \delta_t + \delta_c + \epsilon_c t \quad (15)$$

$$\ln p^* = \beta_1 \text{tariff} + \beta_2 \text{tariff} * \text{HQ} + \delta_t + \delta_c + \epsilon_c t \quad (16)$$

4.4 Quality Variations and quasi-Metzler variation

We can evaluate the change in efficiency in import in the U.S. demand through different channels of the quasi-Metzler :

- New firms entering
- New models being proposed to the market $\longrightarrow$ more efficient and/or more pricy
- Persistence of some models on the market: what are the characteristics of models that stay on the market, those with the lowest price elasticity, the highest quality? (Kaplan-Meier, Cox Model), market share by segment of the market.

A Mathematical Appendix

A.1 Consumer Demand

Development of the calculus for (??):

$$\begin{aligned}
\gamma q_i^c &= \alpha + z_i^l - p_i^l - \eta Q_c^l \\
\iff q_i^c &= \frac{1}{\gamma} \left[\alpha + z_i^l - p_i^l - \eta \int_{i \in \Omega} q_i^c di \right] \\
\iff q_i^c &= \frac{1}{\gamma} \left[\alpha + z_i^l - p_i^l - \eta N^l \bar{q} \right] \\
\iff q_i^c &= \frac{1}{\gamma} \left[\alpha + z_i^l - p_i^l - \eta N^l \left(\frac{1}{\gamma + \eta N^l} [\alpha - \bar{z}^l + \bar{p}] \right) \right] \\
\iff q_i^c &= \frac{1}{\gamma} \frac{\alpha \gamma}{\gamma + \eta N^l} + \frac{1}{\gamma} z_i^l - \frac{1}{\gamma} p_i^l - \frac{1}{\gamma} \frac{N^l \eta}{\gamma + \eta N^l} \bar{z}^l + \frac{1}{\gamma} \frac{N^l \eta}{\gamma + \eta N^l} \bar{p}^l \\
\iff q_i^c &= \frac{\alpha}{\gamma + \eta N} + \frac{1}{\gamma} z_i^l - \frac{1}{\gamma} p_i^l - \frac{1}{\gamma} \frac{N^l \eta}{\gamma + \eta N^l} \bar{z}^l + \frac{1}{\gamma} \frac{N^l \eta}{\gamma + \eta N^l} \bar{p}^l \\
\iff q^c \equiv L^l q_i^c &= \frac{L^l \alpha}{\gamma + \eta N^l} + \frac{L^l}{\gamma} z_i^l - \frac{L^l}{\gamma} p_i^l - \frac{L^l}{\gamma} \frac{N^l \eta}{\gamma + \eta N^l} \bar{z}^l + \frac{L^l}{\gamma} \frac{N^l \eta}{\gamma + \eta N^l} \bar{p}^l
\end{aligned}$$

with $\bar{q} = \frac{1}{N} \int_{i \in \Omega^*} q_i^c di$, $\bar{p} = \frac{1}{N} \int_{i \in \Omega^*} p_i di$, $\bar{z} = \frac{1}{N} \int_{i \in \Omega^*} z_i di$ and $\Omega^* \subset \Omega$ the subset of consumed differentiated goods.

A.2 Firms

Development of the calculus for (??). We want to derive from the profit function the profit-maximizing equations for quantity, price, and quality.

First we take the derivative of π with respect to p , to obtain the optimal quantity function:

$$\begin{aligned}
\frac{\partial \pi^{fh}}{\partial p^{fh}} &= q^{fh} + \frac{\partial q^{fh}}{\partial p^{fh}} p^{fh} - \delta \tau^h c \frac{\partial q^{fh}}{\partial p^{fh}} - \tau^h \theta z^{fh} \frac{\partial q^{fh}}{\partial p^{fh}} \\
q^{fh} &= \frac{L^h}{\gamma} (p^{fh} - \delta \tau^h c - \tau^h \theta z)
\end{aligned}$$

Then we use (??) when the firm produces at the threshold price p_{max} for which the demand is $q(p_{max}) = 0$ and $z = 0$:

$$\begin{aligned}
0 &= \frac{L^h \alpha}{\gamma + \eta N^h} - \frac{L^h}{\gamma} p_{max} - \frac{L^h}{\gamma} \frac{N^h \eta}{\gamma + \eta N^h} \bar{z}^h + \frac{L^h}{\gamma} \frac{N^h \eta}{\gamma + \eta N^h} \bar{p}^h \\
\iff \frac{L^h}{\gamma} p_{max} &= \frac{L^h \alpha}{\gamma + \eta N^h} - \frac{L^h}{\gamma} \frac{N^h \eta}{\gamma + \eta N^h} \bar{z}^h + \frac{L^h}{\gamma} \frac{N^h \eta}{\gamma + \eta N^h} \bar{p}^h
\end{aligned}$$

Then we plug $\frac{L^h}{\gamma} p_{max}$ in (??) and find :

$$\begin{aligned}
q^{fh} &= \underbrace{\frac{L^h \alpha}{\gamma + \eta N^h} - \frac{L^h}{\gamma} \frac{N^h \eta}{\gamma + \eta N^h} \bar{z}^h + \frac{L^h}{\gamma} \frac{N^h \eta}{\gamma + \eta N^h} \bar{p}^h}_{\frac{L^h}{\gamma} p_{max}} + \frac{L^h}{\gamma} z_i^{fh} - \frac{L^h}{\gamma} p_i^{fh} \\
\iff q^{fh} &= \frac{L^h}{\gamma} p_{max} + \frac{L^h}{\gamma} z_i^{fh} - \frac{L^h}{\gamma} p_i^{fh}
\end{aligned}$$

Finally we can equate with $q^{fh} = \frac{L^h}{\gamma} (p^{fh} - \delta \tau^h c - \tau^h \theta z)$:

$$\begin{aligned}
\frac{L^h}{\gamma} (p^{fh} - \delta \tau^h c - \tau^h \theta z^{fh}) &= \frac{L^h}{\gamma} p_{max} + \frac{L^h}{\gamma} z_i^{fh} - \frac{L^h}{\gamma} p_i^{fh} \\
\iff p^{fh} - \delta \tau^h c - \tau^h \theta z^{fh} &= p_{max} + z_i^{fh} - p_i^{fh} \\
\iff p^{fh} &= \frac{1}{2} \left[\underbrace{p_{max}}_{\delta \tau^h c^{fh}} + \delta \tau^h c + (1 + \tau^h \theta) z^{fh} \right] \\
\iff p^{fh} &= \frac{1}{2} [\delta \tau^h (c^{fh} + c) + (1 + \tau^h \theta) z^{fh}] \\
\iff p^{fh} &= \frac{1}{2} \delta \tau^h (c^{fh} + c) + \frac{1}{2} (1 + \tau^h \theta) z^{fh}
\end{aligned}$$

With c^{fh} the cost cutoff for exporting to the h market above which demand is 0. We deduce p_{max} as the maximum price for which demand is negative as the transport and tariff inclusive price: $p_{max} = \delta \tau^h c^{fh}$.

Hence we can rewrite the profit-maximizing price, quantity and profit functions as:

$$\begin{aligned}
p^{fh} &= \frac{1}{2} \delta \tau^h (c^{fh} + c) + \frac{1}{2} (1 + \tau^h \theta) z^{fh} \\
q^{fh} &= \frac{L^h}{2\gamma} [\delta \tau^h (c^{fh} - c) + (1 - \tau^h \theta) z^{fh}] \\
\pi^{fh} &= \frac{L^h}{4\gamma} [\delta \tau^h (c^{fh} - c) + (1 - \theta \tau^h) z^{fh}]^2 - \theta (z^{fh})^2
\end{aligned}$$

For the quality function we just take the derivative of the profit function with respect to z :

$$\begin{aligned}
& \frac{\partial \pi^{fh}}{\partial z^{fh}} = 0 \\
\iff & L^h [\delta \tau^h (c^{fh} - c) + (1 - \theta \tau^h) z^{fh}] (1 - \theta \tau^h) = 4\gamma \theta \tau^h z^{fh} \\
& \iff L^h (1 - \theta \tau^h) \delta \tau^h (c^{fh} - c) = z^{fh} [4\gamma \theta \tau^h - L^h (1 - \theta \tau^h)^2] \\
& \iff \frac{L^h (1 - \theta \tau^h)}{4\gamma \theta \tau^h - L^h (1 - \theta \tau^h)^2} \delta \tau^h (c^{fh} - c) = z^{fh} \\
& \iff \lambda^{fh} \delta \tau^h (c^{fh} - c) = z^{fh}
\end{aligned}$$

Therefore, the final optimal quality functions are:

$$\begin{aligned}
z^{fh} &= \lambda^{fh} \delta \tau^h (c^{fh} - c) \\
z^{hh} &= \lambda^{hh} (c^{hh} - c)
\end{aligned}$$

With $\lambda^{fh} = \frac{L^h (1 - \theta \tau^h)}{4\gamma \theta \tau^h - L^h (1 - \theta \tau^h)^2}$ and $\lambda^{hh} = \frac{L^h (1 - \theta)}{4\gamma \theta - L^h (1 - \theta)^2}$ defining the "quality scope" of good exporting from f to h and from home firms h for *Home* markets.

And finally, by plugging this optimal quality choice function in our previous equations we obtain :

$$\begin{aligned}
p^{fh} &= \frac{1}{2} \delta \tau^h (c^{fh} + c) + \frac{1}{2} (1 + \tau^h \theta) \lambda^{fh} \delta \tau^h (c^{fh} - c) \\
q^{fh} &= \frac{L^h}{2\gamma} [\delta \tau^h (c^{fh} - c) + (1 - \tau^h \theta) \lambda^{fh} \delta \tau^h (c^{fh} - c)] \\
\pi^{fh} &= \frac{\delta^2 L^h \tau^h}{4\gamma} [1 + (1 - \theta \tau^h) \lambda^{fh}] (c^{fh} - c)^2
\end{aligned}$$

For home firm we just need to set $\delta = \tau = 1$:

$$\begin{aligned}
p^{hh} &= \frac{1}{2} (c^{hh} + c) + \frac{1}{2} (1 + \theta) \lambda^{hh} (c^{hh} - c) \\
q^{hh} &= \frac{L^h}{2\gamma} [(c^{hh} - c) + (1 - \theta) \lambda^{hh} (c^{hh} - c)] \\
\pi^{hh} &= \frac{L^h}{4\gamma} [1 + (1 - \theta) \lambda^{hh}] (c^{hh} - c)^2
\end{aligned}$$

A.3 Market Equilibrium

We can rewrite (??) as :

$$f_E = \int_0^{c^{fh}} \frac{\delta^2 L^h \tau^h}{4\gamma} [1 + (1 - \theta \tau^h) \lambda^{fh}] (c^{fh} - c)^2 dG(c) + \int_0^{c^{ff}} \frac{L^h}{4\gamma} [1 + (1 - \theta) \lambda^{ff}] (c^{ff} - c)^2 dG(c)$$

We will only consider the profit made in h since the reasoning for the profit made at home (i.e. f , since we take *Foreign* point of view) is similar to the exception of $\tau = \delta = 1$.

We remind that we are considering a Pareto distribution so that $G(c) = (\frac{c}{c_M})^k$ with $c \in [0, c_M]$. To ease the re-expression of our distribution we can derive the value of $G(c)$:

$$\begin{aligned} \frac{d}{dc} G(c) &= g(c) = k c^{k-1} c_M^{-k} \\ dG(c) &= g(c) dc = k c^{k-1} c_M^{-k} dc \end{aligned}$$

We can plug this in our first member:

$$\frac{\delta^2 L^h \tau^h}{4\gamma} [1 + (1 - \theta \tau^h) \lambda^{fh}] \int_0^{c^{fh}} (c^{fh} - c)^2 k c^{k-1} c_M^{-k} dc$$

For the rest of the calculation we set the following variable $u = \frac{c}{c^{fh}}$ such that :

$$\begin{aligned} u = \frac{c}{c^{fh}} &\iff c = c^{fh} u \text{ and } dc = c^{fh} du \\ \text{and } c^{fh} - c &= (1 - u) c^{fh} \end{aligned}$$

Now we have all the elements to simplify our cost-cutoff functions:

$$\begin{aligned} &\frac{\delta^2 L^h \tau^h}{4\gamma} [1 + (1 - \theta \tau^h) \lambda^{fh}] \int_0^1 (1 - u)^2 (c^{fh})^2 k (c^{fh})^{k-1} u^{k-1} c_M^{-k} c^{fh} du \\ \iff &\frac{\delta^2 L^h \tau^h}{4\gamma} [1 + (1 - \theta \tau^h) \lambda^{fh}] (c^{fh})^{k+2} k c_M^{-k} \underbrace{\int_0^1 (1 - u)^2 u^{k-1} du}_{B(k,3) = \frac{\Gamma(3)\Gamma(k)}{\Gamma(k+3)}} \\ \iff &\frac{\delta^2 L^h \tau^h}{4\gamma} [1 + (1 - \theta \tau^h) \lambda^{fh}] (c^{fh})^{k+2} k c_M^{-k} \frac{2}{(k+2)(k+1)} \\ \iff &\frac{\delta^2 L^h \tau^h}{2\gamma(k+2)(k+1)} [1 + (1 - \theta \tau^h) \lambda^{fh}] (c^{fh})^{k+2} k c_M^{-k} \end{aligned}$$

Symmetrically we have:

$$\frac{L^h}{2\gamma(k+2)(k+1)} [1 + (1 - \theta)\lambda^{ff}] (c^{ff})^{k+2} k c_M^{-k}$$

Then plugging back in our free entry equation we obtain :

$$f_E = \frac{\delta^2 L^h \tau^h}{2\gamma(k+2)(k+1)} [1 + (1 - \theta\tau^h)\lambda^{fh}] (c^{fh})^{k+2} k c_M^{-k} + \frac{L^h}{2\gamma(k+2)(k+1)} [1 + (1 - \theta)\lambda^{ff}] (c^{ff})^{k+2} k c_M^{-k}$$

Which naturally leads to :

$$\begin{cases} \gamma\phi = L^h \delta^{-k} (\tau^h)^{-k-1} [1 + (1 - \theta\tau^h)\lambda^{fh}] (c^{hh})^{k+2} + L^f [1 + (1 - \theta)\lambda^{ff}] (c^{ff})^{k+2} \\ \gamma\phi = L^f \delta^{-k} (\tau^f)^{-k-1} [1 + (1 - \theta\tau^f)\lambda^{fh}] (c^{ff})^{k+2} + L^h [1 + (1 - \theta)\lambda^{hh}] (c^{hh})^{k+2} \end{cases}$$

with $\phi = 2f_E(k+2)(k+1)c_M^k$, $c^{fh} = \frac{c^{hh}}{\delta\tau^h}$ and $c^{ff} = \frac{c^{ff}}{\delta\tau^f}$.

From there we just need to re-express this system of equations in function of c^{hh} and c^{ff} :

$$c^{hh} = \left(\frac{\gamma\phi}{L^h} \frac{[1 + (1 - \theta)\lambda^{ff}] - p^f [1 + (1 - \theta\tau^f)\lambda^{fh}]}{[1 + (1 - \theta)\lambda^{hh}][1 + (1 - \theta)\lambda^{ff}] - p^h p^f [1 + (1 - \theta\tau^h)\lambda^{fh}][1 + (1 - \theta\tau^f)\lambda^{fh}]} \right)^{\frac{1}{k+2}}$$

$$c^{ff} = \left(\frac{\gamma\phi}{L^f} \frac{[1 + (1 - \theta)\lambda^{hh}] - p^h [1 + (1 - \theta\tau^h)\lambda^{fh}]}{[1 + (1 - \theta)\lambda^{hh}][1 + (1 - \theta)\lambda^{ff}] - p^h p^f [1 + (1 - \theta\tau^h)\lambda^{fh}][1 + (1 - \theta\tau^f)\lambda^{fh}]} \right)^{\frac{1}{k+2}}$$

Now that we have the cutoff value we can determine a certain number of aggregate values, such as the average quality, prices, and markup with respect to the cutoff value c^{hh} .

In this section, we develop the calculation of the tariff absorption mechanisms of our model. We develop here the total log derivative of (??).

First, we rewrite (??) such that we have it expressed in function of the cost-cutoff and firm productivity separately:

$$p^* = \frac{\delta}{2} [c^{fh}(1 + \kappa) + c(1 - \kappa)]$$

We differentiate with respect to each key elements: c, c^{fh}, κ

$$\begin{aligned} dp^* &= \frac{\delta}{2} [(1 + \kappa)dc^{fh} + c^{fh}d(1 + \kappa) + (1 - \kappa)dc + cd(1 - \kappa)] \\ \iff dp^* &= \frac{\delta}{2} [(1 + \kappa)dc^{fh} + c^{fh}d\kappa + (1 - \kappa)dc - cd\kappa] \\ \iff dp^* &= \frac{\delta}{2} [(1 + \kappa)dc^{fh} + (1 - \kappa)dc + (c^{fh} - c)d\kappa] \end{aligned}$$

And the log derivation :

$$\begin{aligned}
d \ln(p^*) &= \frac{dp^*}{p^*} \\
\iff d \ln(p^*) &= \frac{\frac{\delta}{2} [(1 + \kappa)dc^{fh} + (1 - \kappa)dc + (c^{fh} - c)d\kappa]}{p^*} \\
\iff d \ln(p^*) &= \frac{\frac{\delta}{2} [(1 + \kappa)c^{fh}d \ln(c^{fh}) + (1 - \kappa)cd \ln(c) + (c^{fh} - c)\kappa d \ln(\kappa)]}{p^*} \\
\iff d \ln(p^*) &= \frac{\frac{\delta}{2} [(1 + \kappa)c^{fh}d \ln(c^{fh}) - (1 - \kappa)cd \ln(\frac{1}{c}) + (c^{fh} - c)\kappa d \ln(\kappa)]}{p^*} \\
\iff d \ln(p^*) &= \frac{\delta}{2} \left[\underbrace{\frac{(1 + \kappa)c^{fh}d \ln(c^{fh})}{p^*}}_{\text{Market Condition}} - \underbrace{\frac{(1 - \kappa)cd \ln(\frac{1}{c})}{p^*}}_{\text{Productivity}} + \underbrace{\frac{(c^{fh} - c)\kappa d \ln(\kappa)}{p^*}}_{\text{Quality}} \right]
\end{aligned}$$

A.4 Tariffs Absorption

In this section, we develop the proof of derivation for the elasticity of prices with respect to tariffs (??).

$$\frac{\partial \ln p^*}{\partial \ln \tau^h} = \frac{\delta}{2} \frac{\partial}{\partial \tau^h} \left[\frac{(1 + \kappa)c^{fh} \ln(c^{fh})}{p^*} - \frac{(1 - \kappa)c \ln(\frac{1}{c})}{p^*} + \frac{(c^{fh} - c)\kappa \ln(\kappa)}{p^*} \right]$$

We derive separately $\frac{\partial \ln(c^{fh})}{\partial \ln \tau^h}$ and $\frac{\partial \ln(1+\kappa)}{\partial \ln \tau^h}$.

$$\frac{\partial \ln(c^{fh})}{\partial \ln \tau^h} =$$

A.5 Aggregation

First, we determine the average prices of exports to H and firms in H selling to their domestic market. We start from our price equations in :

$$\begin{aligned}
p^{fh} &= \frac{1}{2} \delta \tau^h (c^{fh} + c) + \frac{1}{2} (1 + \tau^h \theta) \lambda^{fh} \delta \tau^h (c^{fh} - c) \\
p^{hh} &= \frac{1}{2} (c^{hh} + c) + \frac{1}{2} (1 + \theta) \lambda^{hh} (c^{hh} - c)
\end{aligned}$$

We take the average of those prices :

$$\begin{aligned}
\int_0^{c^{fh}} p^{fh} &= \int_0^{c^{fh}} \frac{1}{2} \delta \tau^h (c^{fh} + c) + \frac{1}{2} (1 + \tau^h \theta) \lambda^{fh} \delta \tau^h (c^{fh} - c) dc \\
\iff \bar{p}^{fh} &= \frac{1}{2} \delta \tau^h \int_0^{c^{fh}} (c^{fh} + c) dc + \frac{1}{2} (1 + \tau^h \theta) \lambda^{fh} \delta \tau^h \int_0^{c^{fh}} (c^{fh} - c) dc \\
\iff \bar{p}^{fh} &= \frac{1}{2} \delta \tau^h c^{fh} \frac{2k+1}{k+1} + \frac{1}{2} (1 + \tau^h \theta) \lambda^{fh} \delta \tau^h \frac{c^{fh}}{k+1} \\
\iff \bar{p}^{fh} &= c^{fh} \left[\frac{(2k+1) \tau^h \delta + (1 + \tau^h \theta) \lambda^{fh} \delta \tau^h}{2k+2} \right] \\
\iff \bar{p}^{fh} &= c^{hh} \left[\frac{(2k+1) + (1 + \tau^h \theta) \lambda^{fh}}{2k+2} \right]
\end{aligned}$$

The third equation comes from the Pareto distribution of c and the rule of integration. The last equation comes from $c^{fh} = \frac{c^{hh}}{\delta \tau^h}$ and we factor out $\delta \tau^h$ from the numerator.

$$\begin{aligned}
&\frac{1}{2} (1 + \tau^h \theta) \lambda^{fh} \delta \tau^h \int_0^{c^{fh}} (c^{fh} - c) dc \\
\iff &\frac{1}{2} (1 + \tau^h \theta) \lambda^{fh} \delta \tau^h \int_0^{c^{fh}} c^{fh} dc - \int_0^{c^{fh}} c dc \\
\iff &\frac{1}{2} (1 + \tau^h \theta) \lambda^{fh} \delta \tau^h c^{fh} - \underbrace{\frac{k c^{fh}}{k+1}}_{\text{Mean of a Pareto distribution}} \\
\iff &\frac{1}{2} (1 + \tau^h \theta) \lambda^{fh} \delta \tau^h \frac{c^{fh}}{k+1}
\end{aligned}$$

We follow the same logic to find $\bar{p}^{hh}$ and find:

$$\begin{aligned}
\bar{p}^{hh} &= \frac{1}{2} (c^{hh} + \bar{c}) + \frac{1}{2} (1 + \theta) \lambda^{hh} (c^{hh} - \bar{c}) \text{ with } \bar{c} = \frac{c^{hh} k}{k+1} \\
\iff \bar{p}^{hh} &= \frac{1}{2} \left(c^{hh} + \frac{c^{hh} k}{k+1} \right) + \frac{1}{2} (1 + \theta) \lambda^{hh} \left(c^{hh} - \frac{c^{hh} k}{k+1} \right) \\
\iff \bar{p}^{hh} &= c^{hh} \left(\frac{2k+1 + (1 + \theta) \lambda^{hh}}{2k+2} \right)
\end{aligned}$$

Then, we can estimate the average price applied in the market h as being the weighted prices of *home* and *foreign* companies:

$$\begin{aligned}
\bar{p}_H &= \frac{N_H^f}{N_H} \bar{p}^{fh} + \frac{N_H^h}{N_H} \bar{p}^{hh} \\
\iff \bar{p}_H &= \frac{N_H^f}{N_H} c^{hh} \left[\frac{(2k+1) + (1 + \tau^h \theta) \lambda^{fh}}{2k+2} \right] + \frac{N_H^h}{N_H} c^{hh} \left[\frac{2k+1 + (1 + \theta) \lambda^{hh}}{2k+2} \right] \\
\iff \bar{p}_H &= \frac{c^{hh}}{2k+2} \left[\frac{N_H^f}{N_H} ((2k+1) + (1 + \tau^h \theta) \lambda^{fh}) \right] + \left[\frac{N_H^h}{N_H} (2k+1 + (1 + \theta) \lambda^{hh}) \right] \\
\iff \bar{p}_H &= \frac{c^{hh}}{2k+2} \left[\frac{N_H^h + N_H^f}{N_H} (2k+1) + \frac{N_H^f}{N_H} (1 + \tau^h \theta) \lambda^{fh} + \frac{N_H^h}{N_H} (1 + \theta) \lambda^{hh} \right] \text{ with } \frac{N_H^h + N_H^f}{N_H} = 1 \\
\iff \bar{p}_H &= \frac{c^{hh}}{2k+2} \left[(2k+1) + \frac{N_H^f}{N_H} (1 + \tau^h \theta) \lambda^{fh} + \frac{N_H^h}{N_H} (1 + \theta) \lambda^{hh} \right]
\end{aligned}$$

Hence we see that average prices increase with c^{hh} . From here we can develop further aggregate measures of quality and markups:

$$\begin{aligned}
\bar{z}^{fh} &= \lambda^{fh} \delta \tau^h (c^{fh} - \bar{c}) \\
\iff \bar{z}^{fh} &= \lambda^{fh} \delta \tau^h \left(c^{fh} - \frac{c^{fh} k}{k+1} \right) \\
\iff \bar{z}^{fh} &= \lambda^{fh} \delta \tau^h \left(\frac{c^{fh}}{k+1} \right)
\end{aligned}$$

$$\begin{aligned}
\bar{z}^{hh} &= \lambda^{hh} (c^{hh} - \bar{c}) \\
\iff \bar{z}^{hh} &= \lambda^{hh} \left(c^{hh} - \frac{c^{hh} k}{k+1} \right) \\
\iff \bar{z}^{hh} &= \lambda^{hh} \left(\frac{c^{hh}}{k+1} \right)
\end{aligned}$$

$$\begin{aligned}
\bar{z}_H &= \frac{N_H^f}{N_H} \bar{z}^{fh} + \frac{N_H^h}{N_H} \bar{z}^{hh} \\
\bar{z}_H &= \frac{N_H^f}{N_H} \lambda^{fh} \delta \tau^h \left(\frac{c^{fh}}{k+1} \right) + \frac{N_H^h}{N_H} \lambda^{hh} \left(\frac{c^{hh}}{k+1} \right) \\
\bar{z}_H &= \frac{N_H^f}{N_H} \lambda^{fh} \left(\frac{c^{hh}}{k+1} \right) + \frac{N_H^h}{N_H} \lambda^{hh} \left(\frac{c^{hh}}{k+1} \right) \\
\bar{z}_H &= \frac{c^{hh}}{k+1} \underbrace{\left(\frac{N_H^f \lambda^{fh} + N_H^h \lambda^{hh}}{N_H} \right)}_{\bar{\lambda}_H}
\end{aligned}$$

With $\bar{\lambda}_H$ being the *weighted average quality scope* of goods on market h , also coined as the *effective quality scope* (?).

The average markups can be expressed as $\mu = p - c$, bringing then :

$$\begin{aligned}
\bar{\mu}_H^{fh} &= \frac{1}{2} \frac{c^{hh}}{k+1} [1 + (1 + \delta \tau^h \theta) \lambda^{fh}] \\
\bar{\mu}_H^{hh} &= \frac{1}{2} \frac{c^{hh}}{k+1} [1 + (1 + \theta) \lambda^{hh}] \\
\bar{\mu}_H &= \frac{1}{2} \frac{c^{hh}}{k+1} \left[1 + \underbrace{\frac{N_H^f}{N_H} (1 + \theta \delta \tau^h) \lambda^{fh} + \frac{N_H^h}{N_H} (1 + \theta) \lambda^{hh}}_{\bar{\psi}} \right]
\end{aligned}$$

B Data Appendix

In this section, we develop the data manipulation to clean our dataset, build our main variables, the issues we faced, and the choices we made (see ??). We also propose additional tables and exhibits on which we based our reasoning in ??.

B.1 Data Cleaning

This section develops extensively the cleaning approach and the construction of the key variables for our analysis.

B.1.1 Tracking the Sun

Our main data set is *Tracking the Sun* (hereafter TTS), published by the Berkeley Lawrence Lab with publicly collected data across the U.S. It is an extensive database considering the universe of installed solar panels in the U.S. with detailed information about the localization, size, total price of installation, rebate⁶, module manufacturer, installer, presence of integrated micro inverter, the technology used, the efficiency, and the presence of battery.

This comprehensive dataset represents 95% of all installed systems in our period and 56% of the overall U.S. installed capacity (?). It includes only the system installed and connected to the U.S. grid, whether mounted on the roof or on the ground, for residential, community or small-scale utility use. It excludes all systems above 1MW of installed capacity, which corresponds to a large utility-scale solar system. A classical home installation is around 300kW.

To Do

- Describe main variables: installed price, subsidy
- Describe key created variables: price per watt.

Cleaning Procedure The first step of our cleaning procedure has been to clean the dataset of misspelled names of module manufacturers. The brands more largely misspelled are also among the biggest, among which are Hanwha Qcells, Canadian Solar, Panasonic, and Solar Power⁷. This allowed us to retrieve thousands of installed systems otherwise lost.

⁶Excluding the Federal ITC.

⁷Respectively South Korean, Chinese, Japanese and Chinese.

After that, we deflated the prices of installed solar systems and the rebate provided by the CPI value we obtained from the Bureau of Labor Statistics. We took the first year of the period we consider, 2010, as a reference year.

We then merged our data with the *zipcode* which matches zip code to county, and provided information on the real estate market from the 2010 American Community Survey conducted by the Census Bureau.

B.2 Tables

To delimit our analysis we start by isolating firms that represented during our period of analysis 90% of the market share. We adopt the firm-level delimitation against solar panel market share, which we could also use, because the tariff is applied to the firm and transferred to all its products. This is therefore the accurate level of analysis in our case.

Table 4: Market Share of Top 15 Manufacturers (2010-2020)

Manufacturer	Market Share	Country of Origin
Maxeon - Sunpower	0.1485	USA
Hanwha QCells	0.1176	South Korea
LG Electronics inc.	0.1095	South Korea
Trina solar	0.0854	China
REC solar	0.0767	Norway
SolarWorld	0.0663	Germany
Canadian Solar	0.0661	China
Jinko Solar	0.0469	China
Kyocera Solar	0.0363	Japan
Yingli Energy (China)	0.0360	China
Panasonic	0.0274	Japan
Hyundai Energy Solutions co., ltd.	0.0270	South Korea
Longi Green Energy Technology co., ltd.	0.0194	China
Silfab	0.0169	USA
Sharp	0.0106	Japan

Notes: The column represents the market share over the period 2010-2020 operating in the United States. Cumulated, these companies represent 90% of the market.

Data: Calculated from Tracking the Sun.

Then looking at Chinese firms concerned by AD we find that most of them are the same between the first and second wave of AD measures in 2012 and

2014. The AD rate imposed on them is roughly similar. This reinforces our intuition that the first wave of tariffs was inefficient - the second wave being an adjustment measure - and targeted all Chinese firms, collectively accused of unfair competition. We have here a table summarizing for the biggest Chinese firms the tariff change they faced in 2014.

Table 5: Tariff Change for the Largest Chinese Firms

Manufacturer	AD 2012	AD 2015	Difference
Canadian Solar	25.96	24.48	1.48
JA Solar	25.96	24.48	1.48
Jinko Solar	25.96	24.48	1.48
Suntech Power	31.73	29.14	2.59
Trina Solar	18.32	18.32	0.00
Yingli Energy (China)	25.96	24.48	1.48

Sources: This data on Anti-Dumping Tariff have been collected from ? and cross-checked with primary sources from the International Trade Administration.

We observe that the top 5 Chinese firms represent close to 90% of all Chinese-installed solar panels. This highly concentrated situation motivates me to only study this subsample.

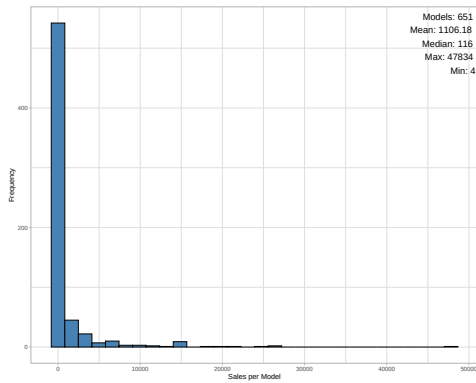
Year	Share Chinese Firms	Top 5 Chinese Firms
2010	0.2639446	0.8935308
2011	0.3643031	0.8885062
2012	0.4417768	0.8653154
2013	0.5036724	0.8216040
2014	0.5344476	0.8258260
2015	0.3688959	0.9308563
2016	0.3484012	0.9196815
2017	0.3212119	0.9619895
2018	0.2693011	0.9298735
2019	0.2347731	0.8192657
2020	0.2331878	0.8955562

Notes: The second column represents the share of installed solar panels by year sourced from a Chinese firm. The third column represents the ratio of those Chinese solar panels coming from one of the top 5 Chinese firms in the period (all affected by AD).

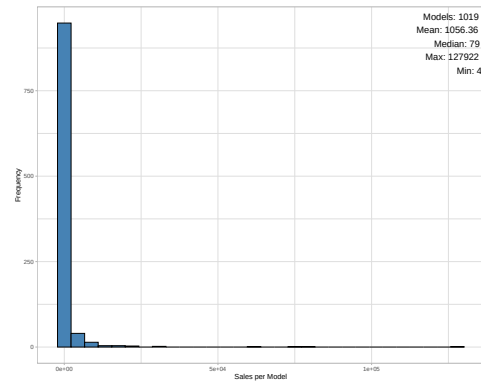
Table 6: Share of Chinese Solar Panels installed each year

The market concentration is not limited to firms but also to models. The distributions of sales by model (see ??) over the decade reveal a very concentrated market with some models making most of the sales. In 2011, just before the first wave of Anti-Dumping tariffs, out of 651 models, 36 accounted for 65% of the sales, 110 for 90%, for a total of only 32 firms out of 101. This highly concentrated configuration supports our choice to adopt a Pareto distribution for productivity draws in ??.

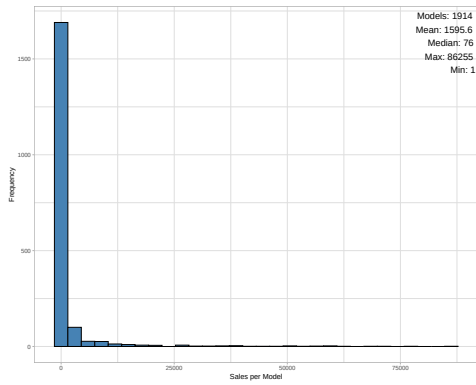
Here we focus on the efficiency criteria of our panel which will be key in determining the quality measures of our panels. We will use these distributions to determine relatively the quality for each wave of tariffs.



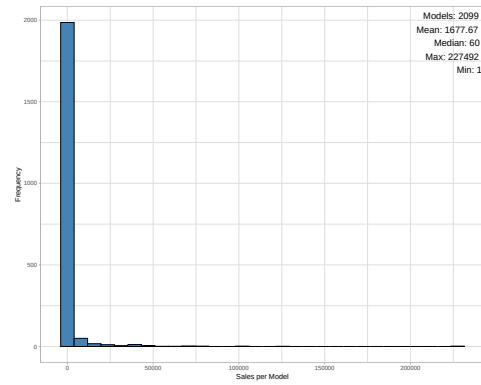
(a) 2011



(b) 2013

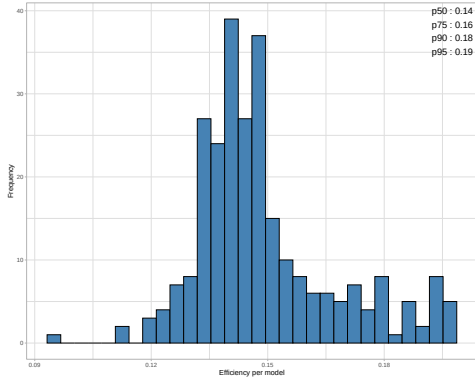


(c) 2017

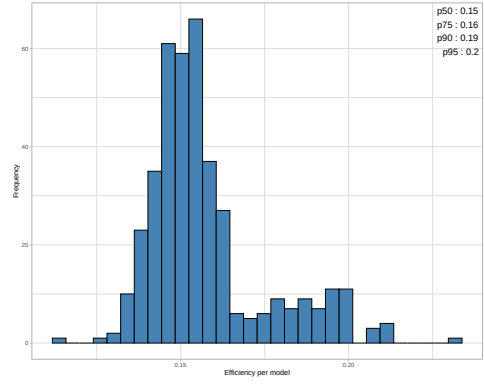


(d) 2019

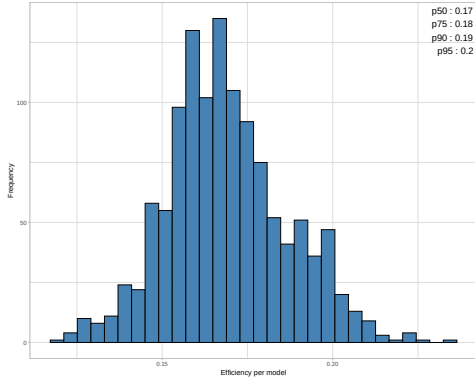
Figure 1: Model Sales Distribution



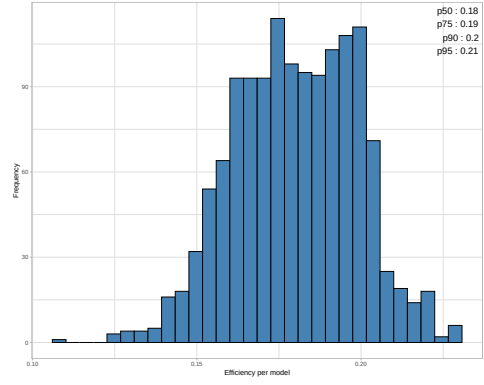
(a) 2011



(b) 2013



(c) 2017



(d) 2019

Figure 2: Model Efficiency Distribution

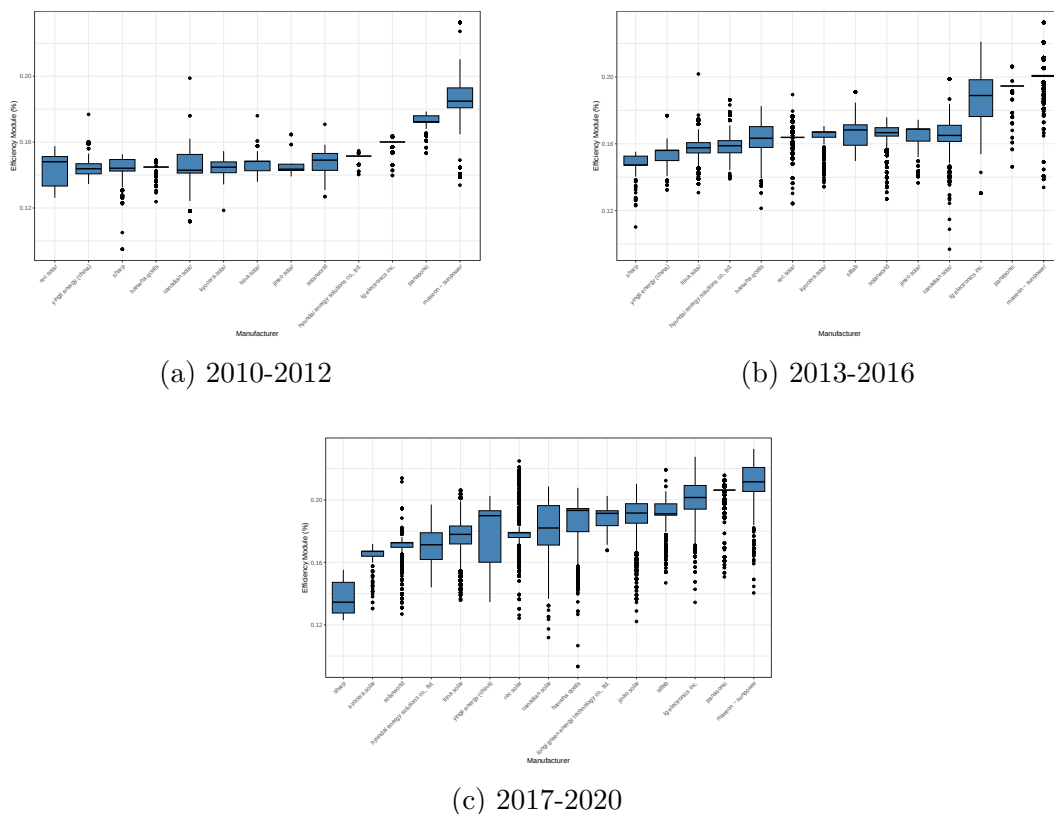


Figure 3: Within firms distribution of solar panels quality.

Interestingly, ?? reveals that for the periods of the anti-dumping tariffs (2010-2012, 2013-2016) we can see an increase in the average level of product delivered, but very small vertical differentiation of product. The third tariff moment introduced is correlated with greater vertical differentiation between and within firms that have introduced more differentiated products. Along with ?? we have some descriptive proof of the effect of tariff on the quality scope of available protected in the protected markets. Indeed, the model predicts that an increase in tariff would decrease the scope for quality by reducing the market access to higher-quality goods. Given the fact the solar market in the U.S. is largely composed of imports, this descriptive effect of tariffs would not be surprising. All exporters to the U.S. need to shift up the quality to maintain themselves on the market like ?? seems to show.

Finally, as a robustness check for quality, and to capture tacit knowledge from brands and panels quality that technical specification cannot capture, we use the analysis of independent expert of [Solar Review](#). They classify the market based on their knowledge of brand quality (i.e. warranty length and

applicability, manufacturing quality, and overall innovation). According to them, LG, Panasonic, and SunPower are *premium* brands for the reliability of their product and innovative content. Tier 1 Chinese brands such as LONGi Solar, Canadian Solar (actually a Chinese company), Trina Solar, Jinko Solar, GCL Group, Risen Energy have slightly less efficient solar panels but are considered high quality given the reliability of their products. We complement it by a review from [PVEL](#), a leading company in evaluating the quality of solar panels. They submit them to a series of test such as :

To Do

- Complete the analysis from PVEL evaluation

We show here the regression table for third-party ownership (i.e. leasing) which is known for its unreliable price reporting (??) and market power over customers (?).

Table 7: Price response to premium panels and installations for TPO systems

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Premium Panels	-0.205 (0.130) [0.166]	-0.143 (0.023) [<0.001]	-0.019 (0.034) [0.591]	-0.082 (0.022) [0.009]				
Premium Installations					-0.487 (0.087) [0.001]	0.075 (0.037) [0.086]	0.079 (0.014) [0.001]	0.087 (0.021) [0.006]
Chinese Brand	0.302 (0.111) [0.035]	0.148 (0.061) [0.050]	0.066 (0.017) [0.009]	-0.132 (0.031) [0.005]	0.298 (0.134) [0.068]	0.174 (0.076) [0.061]	0.062 (0.046) [0.223]	0.003 (0.039) [0.951]
Premium Panels $\times$ Chinese Brand	0.582 (0.335) [0.133]	0.270 (0.010) [<0.001]	0.209 (0.041) [0.002]	0.219 (0.024) [<0.001]				
Premium Installations $\times$ Chinese Brand					-0.025 (0.203) [0.906]	-0.010 (0.040) [0.808]	0.016 (0.057) [0.782]	-0.008 (0.037) [0.841]
Num.Obs.	400 744	400 744	400 744	400 744	400 744	400 744	400 744	400 744
R2	0.021	0.316	0.591	0.601	0.053	0.315	0.593	0.602
FE: zip_code		Yes	Yes	Yes		Yes	Yes	Yes
FE: year_quarter		Yes	Yes	Yes		Yes	Yes	Yes
FE: installer_name			Yes	Yes			Yes	Yes
FE: module_manufacturer		Yes		Yes		Yes		Yes
Wald-test p-value	0.000	0.000	0.000	0.000	0.000	0.034	0.000	0.000

Figure 4: Distribution of installations by installer for HO and TPO systems



Notes: The figures display the log installation by installer for HO and TPO systems.